Chapter 11 — Typography, Colour and Visual Effects

Layout decides *where* boxes go; this chapter is about how they *look*. Most of a web page is text, so typography — the choice of fonts, sizes, line spacing and line length — does more for readability than any other styling. Then come colour and contrast (which decide whether some readers can read at all), and the decorative toolkit: backgrounds and gradients (smooth transitions between colours), borders and rounded corners, shadows, filters (photo-style effects such as blur), blend modes (rules for mixing an element's colours with what is behind it) and image fitting. Each has a performance or accessibility side that is easy to miss.

What you'll learn

#	Topic	You will be able to…
11.1	Fonts	write font stacks (ordered lists of fonts to try), load web fonts (font files downloaded with the page) with `@font-face`, and use a variable font
11.2	Loading fonts well	choose a `font-display` value, preload (fetch early), subset (strip unused characters) and self-host (serve from your own server) fonts
11.3	Text properties	set size, line-height, spacing, decoration, wrapping, truncation (cutting text off with "…") and numerals
11.4	Colour and contrast	compute a WCAG (Web Content Accessibility Guidelines) contrast ratio and derive palettes with `color-mix()` and relative colours
11.5	Backgrounds and gradients	layer images and linear/radial/conic gradients, and size/position them
11.6	Borders, radius and outlines	draw any border, rounded or elliptical corner, and a proper outline
11.7	Shadows	build elevation (the look of a surface floating above the page) with `box-shadow`, use `text-shadow` and `drop-shadow()`
11.8	Filters, blend modes, clipping (cutting an element to a shape) and masks (fading parts of it out)	apply `filter`, `backdrop-filter`, `mix-blend-mode`, `clip-path` and `mask`
11.9	`object-fit` and `object-position`	fit images and video into any box without distortion

As in Chapters 7–10, every example is a page in `code/`, its screenshot in `img/` is made by `code/make_figures.mjs`, and the `NN_*.mjs` scripts use `code/measure.mjs` (a copy of Chapter 10's helper) to read back what Chrome computed — font load status, contrast ratios, even individual **pixel colours** from a screenshot, to check the maths of gradients and blend modes. The one web font used, **Inter**, is in `code/media/` with its SIL Open Font License (a free licence that allows using and redistributing the font).

11.1 Fonts

What it is

A **font family** is a designed set of letter shapes (**glyphs**), usually in several **weights** (thin → black) and **styles** (normal, italic). CSS picks one with `font-family` and the related properties:

```css
body {
  font-family: "Inter", system-ui, -apple-system, "Segoe UI", Roboto, sans-serif;
  font-size: 1rem;           /* 16px by default */
  font-weight: 400;          /* 100–900; normal = 400, bold = 700 */
  font-style: normal;        /* italic | oblique */
  line-height: 1.5;
}
/* shorthand: [style] [variant] [weight] [stretch] size[/line-height] family  (size and family required) */
h1 { font: italic 700 2rem/1.2 Georgia, serif; }
```

Font stacks and generic families

`font-family` is a **fallback list**. For every character the browser tries each family in order and uses the first one that is available *and* has that glyph. Always end with a **generic family**, which every browser maps to something installed:

Generic	Looks like	Typical mapping
`serif`	strokes with small "feet"	Times New Roman
`sans-serif`	no feet	Helvetica / Arial
`monospace`	every character equally wide	Menlo, Consolas, Courier
`system-ui`	the operating system's UI font	San Francisco (Apple), Segoe UI (Windows), Roboto (Android)
`cursive`, `fantasy`	script / decorative	varies wildly — avoid for body text
`ui-serif`, `ui-sans-serif`, `ui-monospace`, `ui-rounded`	the OS's UI variants	only some platforms (Apple mainly)

A **system font stack** (`system-ui, …, sans-serif`) costs zero downloads and looks native (like the operating system's own apps) — GitHub and many design systems (a team's shared set of styles and components) use one. Family names with spaces should be quoted; generic names must *not* be quoted (`"serif"` means a font literally named "serif").

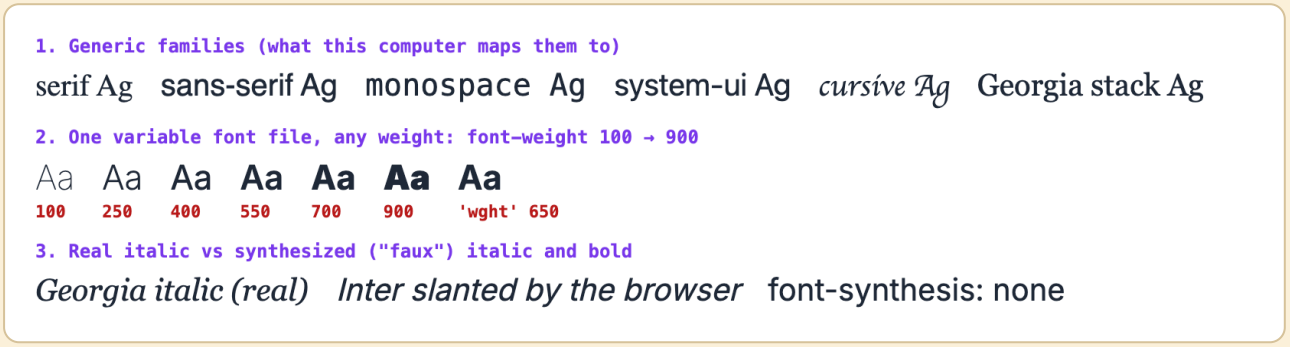


Figure 11.1 — `code/01_fonts.html`: generic families on this Mac, one variable font file at seven weights, and real vs synthesized italic.

`@font-face`: using your own font

```
@font-face {
  font-family: "Inter Var";                                  /* the name YOU choose */
  src: url("media/inter-latin-wght-normal.woff2") format("woff2");
  font-weight: 100 900;                                      /* this file covers a range */
  font-style: normal;
  font-display: swap;                                        /* 11.2 */
}
.inter { font-family: "Inter Var", system-ui, sans-serif; }
```

How it works:

1. `@font-face` only *declares* a font; nothing downloads yet.
2. When the browser lays out text that needs that family (with a matching weight/style), it **downloads the file** — lazily.
3. For each weight/style request, the browser picks the `@font-face` rule whose descriptors (the property-like settings inside `@font-face`: `font-weight`, `font-style`, `font-stretch`) match best. With static fonts (ordinary fonts with one fixed weight and style per file) you write one rule per file (Regular 400, Bold 700, Italic…) all sharing one `font-family` name.
4. If no face matches a requested bold or italic, the browser **synthesizes** one: it smears the glyphs for bold and slants them for italic. `font-synthesis: none` turns this off (see the last sample in Figure 11.1 — it stays upright instead of faking italic).

Font formats

Format	Use
WOFF2	The only one you need today: compressed with Brotli (a modern compression method), supported by every current browser
WOFF	Older compressed format; only for very old browsers
TTF / OTF	Desktop formats; larger; avoid on the web
EOT, SVG fonts	Obsolete

Variable fonts

A **variable font** stores one or more **axes** of variation in a single file, so every weight between 100 and 900 (not just the nine standard steps) comes from one download. Registered axes and the CSS properties that drive them:

Axis tag	Meaning	Prefer this property
`wght`	weight	`font-weight: 550`
`wdth`	width	`font-stretch: 85%`
`ital`	italic on/off	`font-style: italic`
`slnt`	slant angle	`font-style: oblique 10deg`
`opsz`	optical size (letter shapes tuned for small vs large text)	`font-optical-sizing: auto` (default)

`font-variation-settings: "wght" 650, "GRAD" 50` sets axes directly — needed for **custom axes** (uppercase tags such as `GRAD`, grade) but it overrides the higher-level properties and isn't inherited in a composable way (a child setting one axis wipes out the axes its parent set), so use the normal

properties where one exists. Variable fonts are Baseline widely available (supported in all major browsers; since 2018).

`code/01_fonts.mjs` confirms the one file serves the whole range and really changes the outlines:

```
Inter Var weight=100 900 status=loaded display=swap
document.fonts.check("700 22px Inter Var"): true
"Hamburgefonstiv" at 40px, width by weight -> 100: 307.5px, 400: 325.2px, 700: 340.5px, 900: 353.5px
font-synthesis on the last sample: none
```

("Hamburgefonstiv" is a classic type-testing word that contains most letter shapes.) Heavier weights are wider — these are genuinely different glyphs, not a smeared regular.

Trade-offs

Option	+	−
System font stack	0 bytes, instant, native feel	Looks different on each OS; less brand identity
One variable web font	Any weight from one file (~50 KB here for Latin)	One larger file; must manage loading (11.2)
Several static web fonts	Each file small	Every extra weight/style is another request

Common mistakes

- No generic family at the end — if the web font fails, you get the browser default (often Times).
- Asking for `font-weight: 600` when only 400 and 700 files exist — the browser picks 700 (the nearest heavier face); `font-weight: bold` on a family with no bold face → faux bold (fake bold, made by the browser thickening the regular letters).
- Loading five weights "just in case". Check which weights the design actually uses.
- Quoting generic names: `font-family: "sans-serif"` is a (missing) font called sans-serif.

Self-check

1. Why end a stack with `sans-serif`? → It always resolves to an installed font.
2. What does `font-weight: 100 900` inside `@font-face` mean? → This one file supports every weight in that range (a variable font).
3. What is faux bold and how do you prevent it? → Browser-synthesized bold when no bold face exists; provide the face or set `font-synthesis: none`.

11.2 Loading fonts well: `font-display`, preload, subsetting

The problem it solves

A web font is a separate download that usually starts **late** — only after HTML and CSS are parsed and the browser knows the text needs it. Meanwhile the browser must decide what to show. Two historical failure modes:

- **FOIT** (Flash Of Invisible Text): text is hidden until the font arrives. On a slow network, readers stare at blank space.
- **FOUT** (Flash Of Unstyled Text): text shows in a fallback font, then **swaps** — the reflow (re-calculation of layout) may shift content (hurting **CLS**, Cumulative Layout Shift, a Core Web Vital — one of Google's headline page-experience metrics — Chapter 14).

`font-display`

The `font-display` descriptor chooses the trade-off by setting two periods (CSS Fonts Level 4; Baseline widely available):

1. **Block period** — text is drawn invisibly while the font loads.
2. **Swap period** — text is drawn in the fallback; if the font arrives, it swaps in.
3. **Failure period** — too late: keep the fallback for this page.

Value	Block period	Swap period	Use for
`auto`	browser decides (usually like `block`)		—
`block`	short (spec suggests 3 s)	infinite	icon fonts (fonts whose characters are pictures), where fallback glyphs would be wrong
`swap`	extremely short (≈0–100 ms)	infinite	brand/heading fonts that must appear
`fallback`	extremely short	short (≈3 s)	body text: use the font only if it comes quickly
`optional`	extremely short (≈100 ms)	**none**	body text where stability matters most; the font is used from cache (the browser's store of already-downloaded files) on later pages

`code/02_font_display.html` declares the same font three times with `block`, `swap` and `optional` (a Georgia fallback so you can *see* which is in use). `code/02_font_display.mjs` intercepts the network (catches each request before it goes out) and delays every font download by 1.5 seconds (timings are machine-dependent):

```
t≈301ms  F-block=loading  F-swap=loading  F-optional=loading
t≈3503ms  F-block=loaded  F-swap=loaded  F-optional=loaded
  .block     drawn with: Menlo, Inter (web font)
  .swap      drawn with: Menlo, Inter (web font)
  .optional  drawn with: Georgia, Menlo
```

(Menlo is the monospace font of the red `<code>` labels.) The last line is the interesting one: the `optional` font **did** finish downloading, but its window had closed, so the paragraph stays in Georgia for this page view.

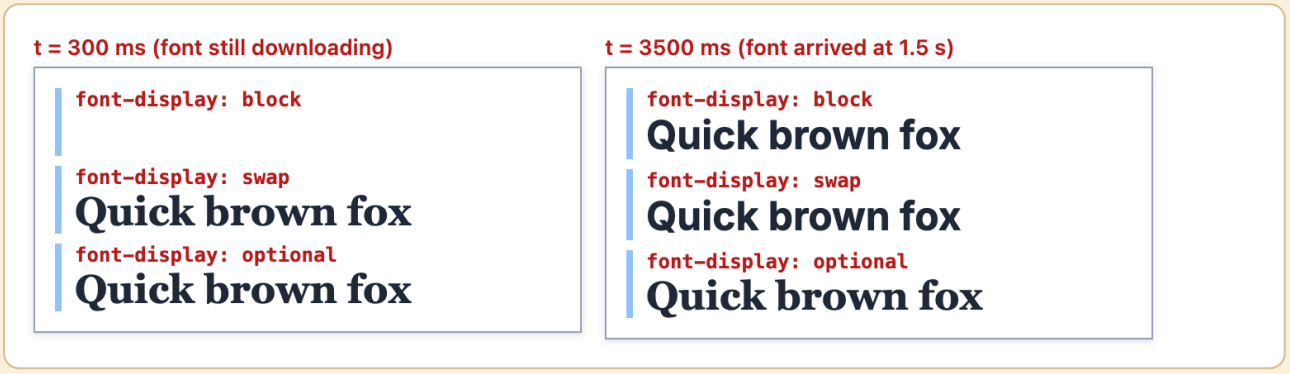


Figure 11.2 — FOIT vs FOUT vs "no swap", captured by delaying the font download.

Making fonts arrive sooner

Technique	How	Why
Preload the critical font	`<link rel="preload" href="/fonts/inter.woff2" as="font" type="font/woff2" crossorigin>`	Starts the download while HTML is parsed instead of after CSS. Font requests are always CORS-mode (CORS, Cross-Origin Resource Sharing, is the set of rules for loading resources across sites), so `crossorigin` is required even for same-origin files — without it the preload is wasted. Preload only 1–2 fonts.
Self-host	Serve the files from your own domain	No extra DNS/TLS connection (no extra address lookup and secure handshake with another server; see System Design Fundamentals); no third-party (another company's server) privacy question (in January 2022 the Munich Regional Court — LG München I, case 3 O 17493/20 — held that loading Google-hosted fonts sent visitors' IP addresses (the numeric network address of their device) to Google without consent, breaching the GDPR (the EU's data-protection law), and awarded the visitor €100)
Subset	Keep only the characters you need; declare `unicode-range: U+0000-00FF, …`	The Inter file here is only the Latin subset (48 KB). Browsers download a `unicode-range` face only if the page uses one of those characters.
Match the fallback's metrics (its letter sizes and line spacing)	`@font-face { font-family: "Inter Fallback"; src: local("Arial"); size-adjust: 107%; }`	The swap then barely moves text, reducing CLS. `size-adjust` works in all current browsers (Safari since 17); `ascent-override` / `descent-override` are not yet supported in Safari.
Fewer files	One variable font instead of 4–6 static ones	Fewer requests
Cache	Long `Cache-Control` max-age (an HTTP header saying how long the browser may reuse the file) with hashed file names (names containing a fingerprint of the content, so a new version gets a new name)	Later pages are instant

From JavaScript

The **CSS Font Loading API** (`document.fonts`; an API is a set of functions a program offers to other code) exposes each face (`status: "unloaded" | "loading" | "loaded" | "error"`), `document.fonts.load('700 1em Inter')` to start a load, and `document.fonts.ready` (a promise — a JavaScript object representing a result that arrives later) — which the chapter scripts await before measuring.

In the interview

30-second answer: "Web fonts load late, so text either goes invisible (FOIT) or shows a fallback then swaps (FOUT) and can shift layout. I choose `font-display`: `swap` for brand text, `optional` or `fallback` for body text, `block` only for icon fonts. I self-host WOFF2, subset with `unicode-range`, preload the one or two critical files with `crossorigin`, use a variable font instead of many weights, and tune the fallback with `size-adjust` to reduce CLS."

Follow-ups:

- *Why does font preload need `crossorigin`?* → Fonts are fetched in CORS mode; a preload without it doesn't match the real request, so the font downloads twice.
- *When is `optional` better than `swap`?* → When layout stability matters more than showing the brand font on the first view; the font is cached and used on the next page.

Self-check

1. What does a user see at 300 ms with `font-display: block` on a slow network? → Invisible text (up to about 3 s).
2. Which value never swaps after the first ~100 ms? → `optional`.
3. Name two ways to reduce font-related layout shift. → `font-display: optional`; metric-matched fallback with `size-adjust`; preloading.

11.3 Text properties

The essentials

Property	Example	Notes
`font-size`	`1.125rem`	Use `rem` for text so user settings work (Chapter 6); fluid sizes with `clamp()` (§10.7)
`line-height`	`1.5`	**Unitless** (see below). Body 1.4–1.6, headings 1.1–1.25
`letter-spacing`	`0.08em`	Small positive values for ALL-CAPS labels; slightly negative for large headings
`word-spacing`	`0.1em`	Rarely needed
`text-align`	`start` | `end` | `center` | `justify`	`start`/`end` follow writing direction; avoid `justify` without hyphenation (rivers of space: gaps lining up down the paragraph)
`text-transform`	`uppercase` | `capitalize`	Keep real capitals in the HTML if they matter; screen readers (software that reads pages aloud) read the source text
`text-indent`	`2em`	First-line indent (books)
`text-decoration`	`underline wavy red`	Longhands: `-line`, `-style` (solid/double/dotted/dashed/wavy), `-color`, `-thickness`
`text-underline-offset`	`0.2em`	Moves the underline away from descenders (the tails of g, p, y)
`white-space`	`nowrap` | `pre` | `pre-wrap`	Controls collapsing of spaces and wrapping
`overflow-wrap`	`anywhere` | `break-word`	Break a long word/URL only if it would overflow (§10.10)
`word-break`	`break-all`	Break between any letters — harsh; mainly for CJK (Chinese, Japanese, Korean) or code
`hyphens`	`auto`	Dictionary hyphenation; needs a `lang` attribute; Baseline since 2023
`text-wrap`	`balance` | `pretty`	Better line breaking (below)
`font-variant-numeric`	`tabular-nums`	Equal-width digits for tables and prices
`max-inline-size`	`65ch`	The **measure** — line length

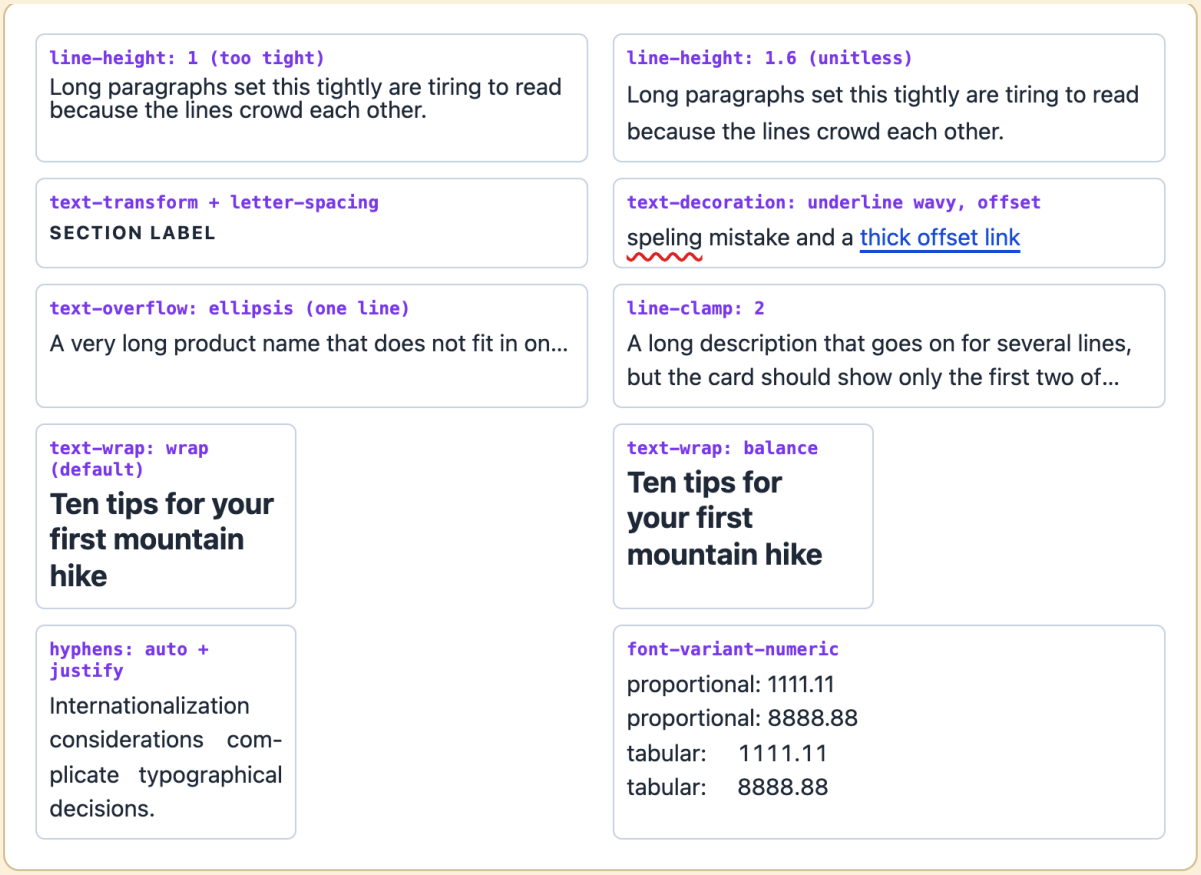


Figure 11.3 — `code/03_text.html`.

Why `line-height` should be unitless

`code/03_text.mjs` puts a 32px child inside a 16px parent with `line-height: 1.5` and `1.5em`:

```
line-height: 1.5   -> child (32px) line box 48px
line-height: 1.5em -> child (32px) computed 24px, line box 24px  (text taller than its line!)
proportional-nums width of "1" 6.73px, of "8" 9.36px
tabular-nums      width of "1" 9.22px, of "8" 9.22px
.caps letter-spacing computed: 0.96px (0.08em of 12px)
.balance text-wrap: balance
```

A unitless value is inherited as a **factor** and re-multiplied by each element's own font size (32 × 1.5 = 48px). A length (`em`, `px`, `%`) is computed **once** on the parent (16 × 1.5 = 24px) and that fixed pixel value is inherited — so bigger text overlaps its neighbours. The same output proves `tabular-nums`: every digit becomes 9.22px wide, so columns of numbers line up.

Line length (the measure)

Comfortable reading needs roughly **45–75 characters per line** (a classic typographic rule of thumb). `max-inline-size: 65ch` (the `ch` unit ≈ the width of "0") keeps paragraphs in that range at any screen size.

Better line breaking: `text-wrap`

- `text-wrap: balance` makes all lines of a short block about equally long — no single orphaned word (a lone word left on the last line) under a long headline. Browsers only balance short blocks (a handful of lines) because it is expensive. **Baseline since May 2024.**
- `text-wrap: pretty` uses a slower algorithm for paragraphs that mainly avoids a very short last line. Supported in Chrome and Safari 26 but not Firefox — **not Baseline**; harmless to add as an enhancement.

Truncation

```
/* One line with "…" — all three are required */
.title { white-space: nowrap; overflow: hidden; text-overflow: ellipsis; }

/* N lines with "…" — the prefixed syntax is the one that works in every browser */
.desc { display: -webkit-box; -webkit-box-orient: vertical; -webkit-line-clamp: 2; overflow: hidden; }
```

The unprefixed `line-clamp` property is specified but not yet implemented by browsers; the `-webkit-` (vendor-prefixed — originally one engine's own extension) version above is interoperable (works the same in every browser). Truncation hides content from sighted users only — make sure the full text is reachable (a details page, or a `title`/tooltip is **not** enough for touch and keyboard users).

A common trap, visible while building Figure 11.3: in a `1fr` grid column, a `nowrap` line makes the **column** grow to fit the text, so no ellipsis appears. The fix is `minmax(0, 1fr)` (§9.10) or `min-width: 0` on the flex/grid item.

Common mistakes

- `line-height` in `px` on `body` — every heading inherits the same fixed spacing.
- Light grey, thin, small text — fails contrast (11.4) and is hard to read.
- Centre-aligning long paragraphs — ragged left edges are hard to scan.
- All-caps paragraphs; `text-transform: uppercase` on long text.
- Removing link underlines in body text: colour alone must not be the only way to identify a link (WCAG 1.4.1) unless the colour contrast with the surrounding text is at least 3:1 *and* there's another cue on hover/focus.

Self-check

1. Parent 20px, `line-height: 1.2em`; child 40px. Child's line-height? → 24px (computed on the parent and inherited) — too tight. With `1.2`: 48px.
2. Three properties needed for a one-line ellipsis? → `white-space: nowrap`, `overflow: hidden`, `text-overflow: ellipsis`.
3. What does `font-variant-numeric: tabular-nums` do? → Makes all digits the same width so numbers align in columns.

11.4 Colour and contrast

Chapter 6 §6.8 introduced colour notations (hex, `rgb()`, `hsl()`, `oklch()`, alpha, `currentColor`). This section is about choosing colours people can read, and deriving palettes from one brand colour.

Contrast ratio — what it is

WCAG defines **contrast ratio** between two colours as `(L1 + 0.05) / (L2 + 0.05)`, where L1 is the **relative luminance** (how bright a colour looks, from 0 for black to 1 for white) of the lighter colour and L2 of the darker. It runs from **1:1** (same colour) to **21:1** (black on white).

How it is computed

`code/contrast.mjs`:

```
// 1. sRGB channel 0-255 -> 0-1 -> linear light (undo the gamma curve)
export const lin = c => { c /= 255; return c <= 0.04045 ? c / 12.92 : ((c + 0.055) / 1.055) ** 2.4; };
// 2. relative luminance: green counts most, blue least (how human eyes weigh them)
export const luminance = ([r, g, b]) => 0.2126 * lin(r) + 0.7152 * lin(g) + 0.0722 * lin(b);
// 3. ratio of the lighter to the darker, each + 0.05 (for screen flare)
export const ratio = (a, b) => { const [hi, lo] = [luminance(a), luminance(b)].sort((x, y) => y - x);
return (hi + 0.05) / (lo + 0.05); };
```

(The printed WCAG 2.0 formula uses 0.03928 as the threshold; newer text corrects it to 0.04045 — for 8-bit colours both give identical results.)

The thresholds (WCAG 2.2)

Criterion	Level	Minimum ratio
1.4.3 Contrast (normal text)	AA	**4.5:1**
1.4.3 Large text (≥ 24px, or ≥ 18.66px bold)	AA	**3:1**
1.4.6 Enhanced contrast	AAA (the highest, strictest WCAG level)	7:1 (large: 4.5:1)
1.4.11 Non-text contrast (icons, input borders, focus rings, chart lines)	AA	**3:1** against adjacent colours

Logos and purely decorative or disabled elements are exempt. See Chapter 3 §3.9 for the rest of WCAG.

Output of `code/04_contrast.mjs` — first by the formula, then using colours the browser computed from `code/04_contrast.html` (the page runs the same formula and prints the result on each swatch):

```
#000000 on #ffffff: 21.00:1  AAA
#767676 on #ffffff: 4.54:1  AA
#777777 on #ffffff: 4.48:1  large/UI only
#1d4ed8 on #ffffff: 6.70:1  AA
#ffffff on #3b82f6: 3.68:1  large/UI only
#e2e8f0 on #0f172a: 14.48:1  AAA
#fcd34d on #ffffff: 1.44:1  FAIL
#fff on var(--brand)                          -> rgb(255, 255, 255) on oklch(0.55 0.2 260)  = 5.02:1 AA
#fff on color-mix(in oklch, brand 70%, black) -> rgb(255, 255, 255) on oklch(0.385 0.14 260)  =
10.04:1 AAA
brand on oklch(from brand 95% 0.03 h)         -> oklch(0.55 0.2 260) on oklch(0.95 0.03 260)  = 4.32:1
large/UI only
#000 on oklch(from brand l c calc(h + 180))   -> rgb(0, 0, 0) on oklch(0.55 0.2 80)  = 4.25:1 large/UI
only
```

(Four of the browser rows, identical to the formula rows above them, are omitted.) Things worth noticing:

- **#767676** is the lightest grey that passes AA on white; one step lighter (#777777) fails.
- White on a typical "blue-500" button (#3b82f6) is only 3.68:1 — fine for large bold text, **not** for normal-size labels. Many design systems use a darker shade for button backgrounds for this reason.
- Yellow text on white is nearly invisible (1.44:1).
- The last row: the complementary colour (the one opposite on the colour wheel) has the **same** OKLCH lightness (55%) as the brand, yet black on it gives 4.25:1 while white on the brand gives 5.02:1. OKLCH lightness is perceptual but it is *not* the WCAG luminance; always check the actual pair.



Figure 11.4 — Contrast computed inside the page from the browser's own colours.

Deriving colours: `color-mix()` and relative colour syntax

Instead of hard-coding every shade, derive them from one token (a named design value stored in a custom property):

```
:root { --brand: oklch(55% 0.2 260); }
.darker     { background: color-mix(in oklch, var(--brand) 70%, black); }   /* 70% brand, 30% black */
.tint       { background: oklch(from var(--brand) 95% 0.03 h); }           /* keep hue, set L and C */
.complement { background: oklch(from var(--brand) l c calc(h + 180)); }    /* rotate the hue */
.faded      { background: rgb(from var(--brand) r g b / 50%); }            /* same colour, half
opacity */
```

- `color-mix(in <space>, A p%, B)` — Baseline since 2023. Mixing in `oklch` or `oklab` gives more even results than `srgb`.
- **Relative colour syntax** `<fn>(from <colour> …)` exposes the source's channels as keywords (`l c h`, `r g b`, `h s l`, `alpha`) that you can keep, replace or compute with `calc()`. Baseline since September 2024 (newly available).

Other colour tools

- `currentColor` — the element's `color` value; great for icons and borders that follow text colour.
- `accent-color: #1d4ed8` — tints checkboxes, radios, range sliders and progress bars. Not yet Baseline (Chrome for Android lacks it), but it degrades gracefully (browsers without it simply show the default colour).
- `color-scheme` and `light-dark()` — §10.5.

Accessibility beyond ratios

- **Never use colour alone** to carry meaning (WCAG 1.4.1): add an icon, text or pattern to "red = error", and underline links in text.
- About 1 in 12 men has some form of colour-vision deficiency (colour blindness); red/green pairs are the classic problem. DevTools (Rendering → Emulate vision deficiencies) simulates them.
- **APCA** (Accessible Perceptual Contrast Algorithm), a newer perceptual contrast method, is being explored for the future WCAG 3; it is not a requirement today. WCAG 2.2's ratio is what audits and laws reference.

In the interview

30-second answer: "WCAG contrast is (L1 + 0.05)/(L2 + 0.05) using relative luminance; AA needs 4.5:1 for body text, 3:1 for large text and for UI components like input borders and focus

rings. I check real pairs with DevTools or a script — lightness in HSL or even OKLCH doesn't guarantee it — and I never rely on colour alone. For palettes I define a brand token in OKLCH and derive shades with `color-mix()` or relative colour syntax."

Self-check

1. Minimum AA ratio for a 14px grey caption? → 4.5:1.
2. For a 3px input border against white? → 3:1 (non-text contrast).
3. What does `oklch(from var(--c) l c calc(h + 180))` produce? → The same lightness and chroma with the hue rotated 180° (the complement).

11.5 Backgrounds and gradients

The background properties

Property	Values	Notes
`background-color`	any colour	Painted under all image layers; always set one as a fallback under images
`background-image`	`url()`, gradients, `image-set()`	Comma-separated **layers**; the **first is on top**
`background-repeat`	`repeat` | `no-repeat` | `repeat-x` | `space` | `round`	Default `repeat`
`background-position`	`center`, `right 10px bottom 10px`, `30% 70%`	Where the image sits
`background-size`	`auto` | `cover` | `contain` | `200px 100px`	`cover` fills and crops; `contain` fits entirely
`background-origin`	`padding-box` | `border-box` | `content-box`	Box that positions the image
`background-clip`	`border-box` | `padding-box` | `content-box` | `text`	Box where painting stops
`background-attachment`	`scroll` | `fixed` | `local`	`fixed` is ignored on iOS and is not Baseline; it also costs repaint performance (the browser must redraw on every scroll)
`background`	shorthand	`url(a.jpg) center / cover no-repeat, #e0f2fe` — `size` goes after `position` with a `/`

Gradients

A **gradient** is an image generated by the browser — no download, crisp at any size, animatable (able to be smoothly animated) only indirectly.

Function	Shape	Example
`linear-gradient()`	along a line	`linear-gradient(135deg, #f59e0b, #ef4444 50%, #7c3aed)`
`radial-gradient()`	circles/ellipses from a point	`radial-gradient(circle at 30% 30%, #fff 0 8%, #60a5fa 40%, #1e3a8a)`
`conic-gradient()`	around a point (like a clock hand)	`conic-gradient(#3b82f6 0 40%, #22c55e 0 70%, #f59e0b 0)` — a pie chart
`repeating-*-gradient()`	the pattern repeats	`repeating-linear-gradient(45deg, #fde68a 0 10px, #f59e0b 10px 20px)` — stripes

Colour stops are `colour [position] [position]`. Two stops at the same position make a **hard edge** (no blending), and a position of `0` after a larger one means "same as the previous stop" — that's how the conic pie chart gets sharp slices. Direction: `to right`, `to top left`, or an angle (`0deg` = up, `90deg` = right).



Figure 11.5 — `code/05_backgrounds.html`.

Interpolation colour space

By default gradients blend in **sRGB** (the standard colour space of ordinary screens), which makes complementary colours pass through a muddy grey. Since June 2024 (Baseline) you can pick the space: `linear-gradient(in oklch to right, blue, yellow)`. `code/05_backgrounds.mjs` screenshots the page and reads the pixel in the middle:

```
.srgb   at  50%: rgb(128, 128, 127)
.oklch  at  50%: rgb(0, 208, 188)
.hard   at  49%: rgb(34, 197, 94)
.hard   at  51%: rgb(229, 231, 235)
```

The sRGB midpoint of blue (0, 0, 255) and yellow (255, 255, 0) is exactly grey (128, 128, 127); in OKLCH the hue travels around the colour wheel instead, through a vivid teal. The two `.hard` samples, 2% apart, are the two pure stop colours — no blend.

Multiple backgrounds — a real use

Text over photos needs contrast. Put a gradient layer **above** the image:

```
.hero {
  background:
    linear-gradient(to top, rgb(0 0 0 / .75), transparent 60%),   /* layer 1: on top */
    url("hero.jpg") center / cover no-repeat;                     /* layer 2: below */
  color: #fff;
}
```

Gradient text

```
.gradient-text { background: linear-gradient(90deg, #2563eb, #db2777);
                 -webkit-background-clip: text; background-clip: text; color: transparent; }
```

Write both lines: web-features (the open dataset behind Baseline labels) still lists unprefixed `background-clip: text` as not Baseline. Check contrast at the gradient's lightest point, and keep a solid `color` fallback in forced-colours mode (an operating-system high-contrast setting that replaces page colours with the user's own).

Common mistakes

- Putting the gradient layer **after** the image — it ends up underneath and invisible.
- No `background-color` under a photo: if the image fails, white text sits on white.
- Meaningful images as CSS backgrounds — no `alt`, not announced; use `<img>` (Chapter 4).
- Huge background photos on mobile — use `image-set()` (§10.8).

Self-check

1. In `background: url(a.png), url(b.png)`, which is on top? → `a.png` (first layer).
2. How do you draw a 50/50 hard split? → `linear-gradient(to right, green 50%, grey 50%)`.
3. Why does blue→yellow look grey in the middle by default? → Interpolation in sRGB averages the channels; use `in oklch`.

11.6 Borders, radius and outlines

Borders

`border: <width> <style> <colour>` on all sides, or per side (`border-left`), or per aspect (`border-width: 1px 2px`). **The style is required** — its initial value is `none` (Chapter 7). Styles: `solid`, `dashed`, `dotted`, `double`, `groove`, `ridge`, `inset`, `outset`, `hidden`, `none`. Borders take up layout space (they're part of the box model). `border-image` can slice an image into a border — rarely used.

`border-radius`

Rounds the corners of the **border box** (and clips the background with it):

Value	Result
`8px`	all four corners
`24px 0 24px 0`	top-left, top-right, bottom-right, bottom-left (clockwise)
`50%` on a square	a circle (`aspect-ratio: 1` keeps it square)
`999px`	a **pill**: a very large radius is reduced to fit, so the ends are exact semicircles at any height
`50% / 30%`	**elliptical** corners: horizontal radii / vertical radii

Percentages are relative to the box's own width (horizontal radius) and height (vertical radius). When adjacent radii would overlap, the browser scales all radii down proportionally. (The `corner-shape` property for squircles (smooth square–circle hybrids) and bevels (cut-off corners) is Chromium-only — only in browsers built on Google's engine — in 2026.)

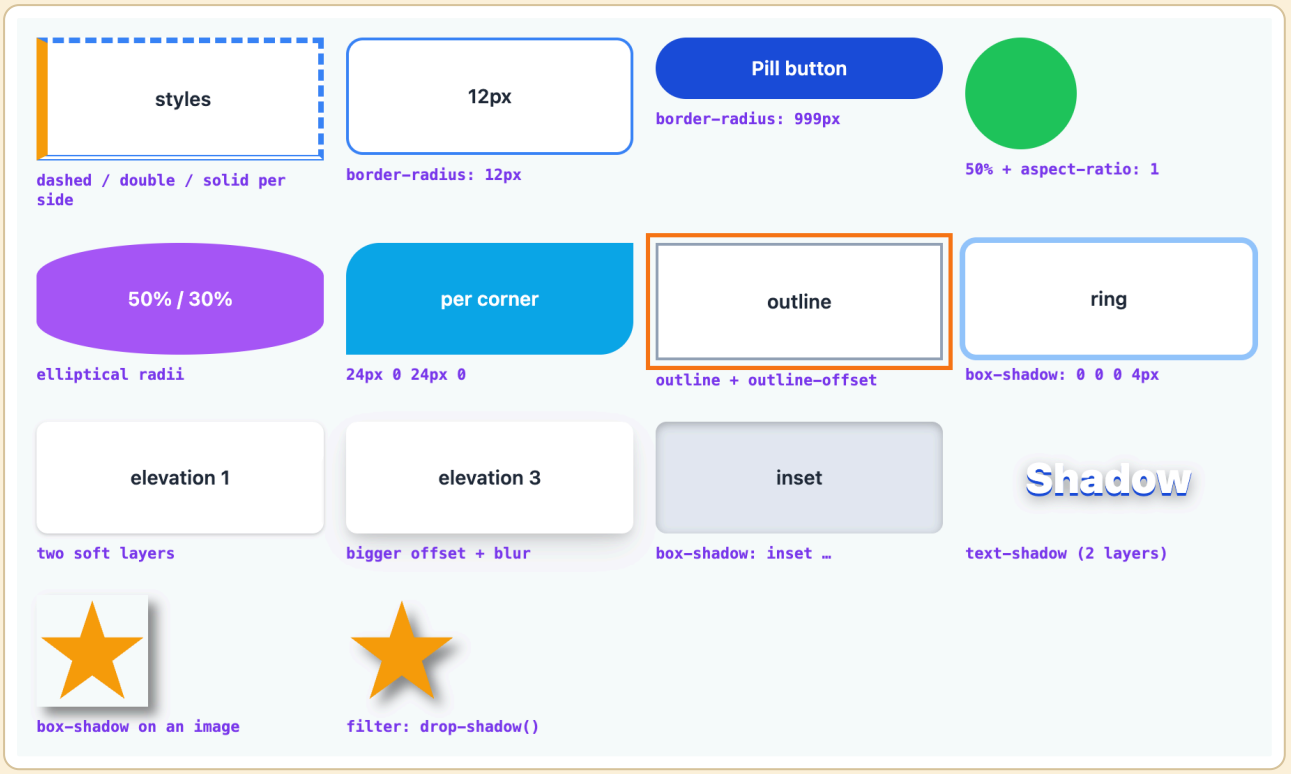


Figure 11.6 — `code/06_borders_shadows.html`.

Outline vs border vs ring

	`border`	`outline`	`box-shadow: 0 0 0 4px` ("ring")
Takes layout space	**yes**	no	no
Per-side control	yes	no	no
Follows `border-radius`	yes	yes (all current browsers since 2023)	yes
Offset	—	`outline-offset` (can be negative)	via extra layers
Shown in forced-colours mode	yes	**yes**	**no** (shadows are removed)
Main use	structure, dividers	**focus indicators**	decorative rings

`code/06_borders_shadows.mjs` confirms that only borders change the layout size (every box has `height: 80px`):

```
.styles   layout box 206x88  border-left 8px  outline none       shadow none
.outline  layout box 206x84  border-left 2px  outline 3px +4px   shadow none
.ring     layout box 206x80  border-left 0px  outline none       shadow rgb(147, 197, 253) 0px 0px 0px
4px
.elev3    layout box 206x80  border-left 0px  outline none       shadow rgba(0, 0, 0, 0.1) 0px 4px 6px
-1px, rgba(0,
.ellipse  border-top-left-radius: 50% 30%
```

This is why focus styles use `outline`: `:focus-visible { outline: 3px solid #2563eb; outline-offset: 2px; }` doesn't shift the layout, and still shows in Windows contrast themes. Never remove the outline without a replacement (Chapter 3 §3.7). A popular pattern is a transparent outline plus a box-shadow ring, so forced-colours mode still gets the outline.

Self-check

1. Why doesn't `border: 2px #333` show? → No style; the default `none` wins.
2. How do you make a pill button at any height? → `border-radius: 999px` (or any value ≥ half the height).
3. Which focus indicator survives forced-colours mode? → `outline` (not `box-shadow`).

11.7 Shadows

`box-shadow`

```
box-shadow: [inset] <offset-x> <offset-y> [<blur>] [<spread>] <colour>, … ;
```

Part	Effect
offset-x / offset-y	Moves the shadow right/down (negative = left/up) — implies the light direction
blur	Softness; 0 = hard edge
spread	Grows (positive) or shrinks (negative) the shadow before blurring
`inset`	Draws inside the box (pressed/sunken look)
multiple, comma-separated	First one on top; layer a tight dark shadow and a large soft one for realistic depth

Elevation systems (Material Design — Google's design system — and most component libraries) express how "high" a surface floats with a set of increasingly large, soft, offset shadows:

```
.elev1 { box-shadow: 0 1px 2px rgb(0 0 0 / .12), 0 1px 3px rgb(0 0 0 / .08); }
.elev3 { box-shadow: 0 4px 6px -1px rgb(0 0 0 / .1), 0 10px 20px -5px rgb(0 0 0 / .15); }
```

The negative spread keeps the large soft shadow from peeking out at the sides. Shadows don't affect layout and are painted outside the box (so `overflow: hidden` on a **parent** clips them).

`text-shadow`

`text-shadow: <x> <y> [<blur>] <colour>, …` — no spread, no inset. Use for legibility on busy images (a soft dark halo) or for display effects; keep body text shadow-free.

`filter: drop-shadow()`

`box-shadow` follows the **box** (a rectangle, or its rounded corners). `filter: drop-shadow(x y blur colour)` follows the element's actual **alpha shape** (the outline of its non-transparent pixels) — the star in Figure 11.6 gets a star-shaped shadow. It has no spread or inset, and costs more to render.

Performance

Large blurred shadows are expensive to paint; **animating** `box-shadow` repaints every frame. To animate elevation, put the bigger shadow on a pseudo-element (a CSS-generated box such as `::after`) and animate its `opacity` instead (Chapter 12).

Self-check

1. What does a negative spread do? → Shrinks the shadow before blur, so it doesn't extend beyond the box's sides.
2. Shadow for a transparent PNG logo that follows its outline? → `filter: drop-shadow(…)`.

11.8 Filters, blend modes, clipping and masks



Figure 11.7 — `code/07_filters_blend.html`.

`filter`

Applies image-processing functions to an element **and everything inside it**, in the order written:

Function	Example	Effect
`blur()`	`blur(3px)`	Gaussian blur (a smooth, even blur)
`brightness()`	`brightness(1.4)`	< 1 darker, > 1 brighter
`contrast()`	`contrast(1.8)`	
`grayscale()`	`grayscale(1)`	0–1 (or %)
`hue-rotate()`	`hue-rotate(120deg)`	Rotates hues
`invert()`	`invert(1)`	
`opacity()`	`opacity(.5)`	Like the `opacity` property
`saturate()`	`saturate(3)`	
`sepia()`	`sepia(1)`	
`drop-shadow()`	11.7	
`url(#svgFilter)`		Any SVG filter

Baseline widely available since 2016. Uses: greyscale logos that gain colour on hover, dimming images in dark mode, disabled states.

`backdrop-filter`

Filters whatever is **behind** the element (the element must be at least partly transparent to see it) — the "frosted glass" of macOS and iOS panels: `background: rgb(255 255 255 / .3); backdrop-filter: blur(6px) saturate(1.4);`. Baseline since September 2024 (Safari 18 dropped the `-webkit-` prefix

requirement). Check text contrast against the *worst* background that can appear behind it, and provide a more opaque fallback for `prefers-reduced-transparency` where supported.

Blend modes

- `mix-blend-mode` — how the element's pixels combine with the content **behind** it.
- `background-blend-mode` — how an element's **own background layers** (images, gradients, colour) combine with each other; used for duotone images (pictures recoloured in two tones).
- `isolation: isolate` — creates a new stacking context (a group of elements layered as one unit) so blending stays inside a group (§7.8).

Main modes, per colour channel with values 0–1 (a = top, b = backdrop): `multiply` = a·b (always darker; white disappears), `screen` = 1 − (1−a)(1−b) (always lighter; black disappears), `overlay`, `darken` / `lighten`, `difference` = |a − b|, `luminosity` / `color` / `hue` / `saturation` (mix HSL-like components). `code/07_filters_blend.mjs` blends #ef4444 over a sky blue and compares the screenshot pixel with the formula:

```
multiply   by hand rgb(94, 40, 63)  browser rgb(94, 40, 63)
screen     by hand rgb(245, 178, 240)  browser rgb(245, 178, 240)
difference by hand rgb(139, 82, 167)  browser rgb(139, 82, 167)
```

Clipping and masking

- `clip-path` hides everything outside a shape: `circle(40% at 50% 50%)`, `ellipse()`, `inset(10px round 8px)`, `polygon(0 0, 100% 0, 100% 75%, 0 100%)`, `path("…")`, or an SVG `url(#clip)`. The shape is hard-edged; clicks outside it pass through. Baseline since 2021 (animatable between shapes with the same number of points).
- `mask-image` uses an image's **alpha** (or luminance) as a soft mask: `mask-image: linear-gradient(to right, #000 40%, transparent)` fades an element out. Unprefixed masks are Baseline since December 2023.

Side effects to remember

A `filter`, `backdrop-filter`, `mix-blend-mode`, `clip-path`, `mask` or `opacity < 1` **creates a stacking context** (§7.8); a non-`none` `filter` or `backdrop-filter` also becomes the **containing block for `position: fixed` descendants** — a "fixed" header inside a filtered parent scrolls with it. Filters and backdrop filters are GPU (graphics chip) work; large blurred areas on low-end phones can drop frames (skip screen updates, so animation stutters).

Self-check

1. Which blend mode makes white disappear? → `multiply`.
2. What's the difference between `filter` and `backdrop-filter`? → `filter` changes the element itself; `backdrop-filter` changes what shows through it.
3. Why might `position: fixed` stop working inside a card? → An ancestor has `filter` / `transform` / `backdrop-filter`, which becomes the containing block.

11.9 `object-fit` and `object-position`

What it is

For **replaced elements** (elements whose content comes from an external file or source, not from CSS) — `<img>`, `<video>`, `<canvas>`, `<iframe>` content — `object-fit` decides how the content fills the element's box when their aspect ratios differ. It's the `background-size` of images. Baseline widely available (2020).

Value	Scale factor	Result
`fill` (default)	separate x and y	Stretched to the box — **distorted**
`contain`	min(box/content) per axis	Whole image visible, **letterboxed** (empty bars fill the leftover space)
`cover`	max(box/content) per axis	Box completely filled, overflow **cropped**
`none`	1	Natural size, centred, cropped
`scale-down`	the smaller of `none` and `contain`	Never enlarges

`object-position` (default `50% 50%`) chooses which part stays visible when cropping or where the letterboxed image sits — like `background-position`.

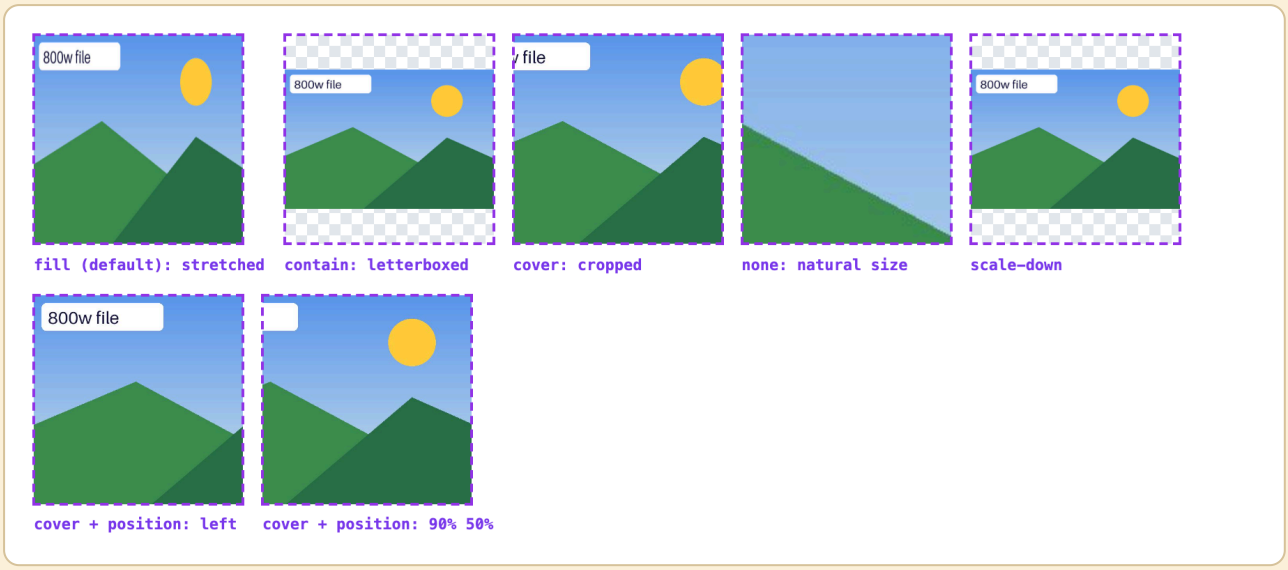


Figure 11.8 — `code/08_object_fit.html`. Dashed purple = the element's box; the checkerboard shows any empty space.

Output of `code/08_object_fit.mjs`, with the maths done by hand from the natural size the browser reports:

```
fill         box 150×150  natural 800×533  object-fit fill       object-position 50% 50%
...
cover-right  box 150×150  natural 800×533  object-fit cover      object-position 90% 50%

contain: scale = min(150/800, 150/533) = 0.1875 -> drawn 150×99.94 (bars above and below)
cover:   scale = max(150/800, 150/533) = 0.2814 -> drawn 225.14×150, 75.14px cropped horizontally
object-position 90% 50% on cover: image x offset = (150 - 225.14) × 0.9 = -67.63px
scale-down: the smaller of none (scale 1) and contain (0.1875) -> same as contain here
```

A percentage position means "align that percentage point of the image with the same percentage point of the box", so the offset is (box − image) × percentage — the same rule as `background-position`.

Real-world use

- **Card thumbnails / avatars** of unpredictable shape: `width: 100%; aspect-ratio: 4 / 3; object-fit: cover;` (and `border-radius: 50%` for avatars).
- **Product photos** that must not be cropped: `object-fit: contain` with a neutral background.
- **Hero video**: `<video autoplay muted loop playsinline>` with `object-fit: cover` to fill a full-width banner.

- **Grid cells**: images inside grid cells don't stretch by default (§9.10); `width: 100%; height: 100%; object-fit: cover` fills them.
- When cropping could cut off the subject (faces), set `object-position` per image, e.g. with an inline `style="object-position: 30% 20%"` from a CMS (content management system — the app editors use to manage site content) focal point (the spot an editor marked as most important).

Common mistakes

- Setting only `width` on an `<img>` and expecting `object-fit` to do anything — the box must have both dimensions (via `height` or `aspect-ratio`) that differ from the image's ratio.
- Using `object-fit` on a `<div>` — it only applies to replaced elements; use `background-size` there.
- Cropping important content (text in images) with `cover`.

Self-check

1. An 800×400 image in a 200×200 box with `cover`: drawn size? → scale = max(0.25, 0.5) = 0.5 → 400×200, 200px cropped horizontally.
2. Same with `contain`? → scale 0.25 → 200×100, 50px bars above and below.
3. `object-position: 0% 50%` keeps which part? → The left edge (vertically centred).

Chapter 11 TL;DR

- **Fonts**: stacks end with a generic family; `system-ui` stack = 0 bytes. `@font-face` + **WOFF2**; one **variable font** (`font-weight: 100 900`) replaces many files — heavier weights really are different glyphs (307.5 → 353.5px wide). Missing faces are **synthesized** unless `font-synthesis: none`.
- **Loading**: `font-display` — `block` (invisible ≤3 s), `swap` (fallback then swap), `fallback` (short swap window), `optional` (≈100 ms, then never — verified: loaded but not used). Preload 1–2 critical fonts **with** `crossorigin`, self-host, subset with `unicode-range`, match fallback metrics with `size-adjust`.
- **Text**: `rem` sizes; **unitless** `line-height` (a length is computed once and inherited — 24px for 32px text); measure 45–75ch; `text-wrap: balance` (Baseline 2024), `pretty` (not Baseline); ellipsis = nowrap + hidden + `text-overflow`; multi-line = `-webkit-line-clamp`; `tabular-nums` for numbers; `hyphens: auto` needs `lang`.
- **Contrast**: (L1+0.05)/(L2+0.05); AA 4.5:1 text, 3:1 large text and UI parts; AAA 7:1. #767676 is the lightest AA grey on white. Never colour alone. Derive shades with `color-mix(in oklch, …)` and relative colours `oklch(from var(--c) l c h)` (Baseline 2024) — and still check each pair.
- **Backgrounds**: layers, first on top; `cover`/`contain`; gradients linear/radial/conic/repeating, hard stops, `in oklch` avoids grey midpoints (measured grey (128,128,127) vs teal).
- **Borders**: style required; `border-radius` %, `999px` pills, `a / b` ellipses. Focus = `outline` (no layout, survives forced colours); rings = `box-shadow: 0 0 0 Npx`.
- **Shadows**: `[inset] x y blur spread colour`, layered for elevation; `text-shadow`; `drop-shadow()` follows alpha. Don't animate shadows directly.
- **Effects**: `filter` (itself), `backdrop-filter` (behind; Baseline 2024), `mix-blend-mode` (multiply = a·b, screen, difference — pixel-verified), `background-blend-mode`, `isolation`, `clip-path`, `mask-image`. They create stacking contexts; filters also capture `position: fixed`.
- **`object-fit`**: fill / contain (min scale) / cover (max scale) / none / scale-down + `object-position` — for img/video in fixed-ratio boxes.

Next: Chapter 12 — Transitions, Transforms and Animation.